



Jenkins

Jenkins – Session 3

Learn CI/CD Tool Jenkins from Scratch

**Integrate GitHub Repository to
Jenkins Project**

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Agenda

- ❑ Integrate Jenkins and GitHub
 - ❑ Create a GitHub Repository
 - ❑ Installation of Git Plugin in Jenkins
 - ❑ Add GitHub Credential in Jenkins
 - ❑ Execute a Job to run GitHub Project
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Integrate Jenkins and GitHub

- ❑ In the real IT world, everyone works on their local machine but at the end of the day, they need to push their complete code into some central repository, such as GitHub repository from where they can run the complete project. GitHub is a Git-based repository host, commonly used for open source projects. GitHub enables code collaboration, hosting, and versioning.
 - ❑ User can integrate Jenkins and GitHub to improve the efficiency of building, testing, and deploying their code.
 - ❑ Integration between Jenkins and GitHub takes on two separate forms:
 - ❖ The ability of Jenkins to pull code from GitHub
 - ❖ The ability of GitHub to trigger Jenkins build jobs
 - ❑ The first feature, namely the ability to pull code from GitHub is made possible through the Jenkins Git plugin. The ability to trigger Jenkins builds when GitHub commits occur is provided through the Jenkins GitHub plugin. Both of these plugins are included as suggested plugins when a default installation of Jenkins CI is performed.
 - ❑ With the Git plugin installed, the Jenkins with GitHub integration is achieved by providing the repository URL and branch name to pull in the build job.
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Create a GitHub Repository

- ❑ Navigate to the link: <https://github.com/>, enter valid credential and sign-in successfully.
 - ❑ Click on the New button.
 - ❑ Enter any repository name and click on "Create Repository". User can also give a description to their repository (optional). By default, a GitHub repository is public which means that anyone can view the contents of this repository whereas in a private repository, user can choose who can view the content. Also, private repository is a paid version.
 - ❑ So now user repository has been successfully created!
 - ❑ Click on the "uploading an existing file" link and select a Java file already created (User can create a sample Java Class if not already done)
 - ❑ Enter the details in Commit changes section and click on the "Commit changes" button.
 - ❑ Note down the GitHub Project Path.
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Installation of Git Plugin in Jenkins

- ❑ Log into Jenkins dashboard through the Jenkins installation path. Unless user have defined a private host, it will be hosted on the localhost at <http://localhost:8080>
 - ❖ Click on the Manage Jenkins link in the left menu bar of Jenkins window
 - ❖ Under the System Configuration section, click on the Manage Plugins link.
 - ❖ Under the Plugin Manager, click on the Available tab and search for the Git plugin.
 - ❖ Select the checkbox in front of the Git plugin and click on the Download now and install after restart button.
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Add GitHub Credential in Jenkins

- ❑ Click on the Manage Jenkins link in the left menu bar of Jenkins window.
 - ❑ Under the System Configuration section, click on the Manage Plugins link.
 - ❑ Under the Plugin Manager, click on the Available tab and search for the Credentials plugin.
 - ❑ Select the checkbox in front of the Credentials plugin and click on the Download now and install after restart button.
 - ❑ Again navigate to Manage Jenkins link from Jenkins window.
 - ❑ Click on the Manage Credentials link under the Security section.
 - ❑ Click on Global Credential link.
 - ❑ Click on Add Credential option to add Git Credentials into Jenkins.
 - ❑ Enter Username & the Password of GitHub account using which user created the GitHub repository.
 - ❑ Don't provide any information in the "ID" field. It will be auto-filled by Jenkins.
 - ❑ Enter any information in the "Description" field as per the user's choice.
 - ❑ Click on the Create button. Then the GitHub repository will be added to Jenkins.
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Execute a Job to run GitHub Project

- ❑ Navigate to Jenkins dashboard
 - ❑ Click on New Item link to create a job on Jenkins
 - ❑ Enter suitable an Item name, select 'Freestyle project' as the option for this new Item and click on OK button.
 - ❑ Describe the project in the description section.
 - ❑ Navigate to the Source Code Management section and select the "Git" option.
 - ❑ Enter GitHub repository link in the Repository URL text box.
 - ❑ Click and select the GitHub credential already add and change the branch specifier to main from master.
 - ❑ Select "Execute Windows batch command" option after clicking on the Add build step dropdown in the Build Steps section.
 - ❑ Enter below command in the text box:
javac ClassName.java
java ClassName
 - ❑ Click on 'Apply' and 'Save'.
 - ❑ Click on 'Build Now' button.
 - ❑ Select Build Number and click on Console Output option to see the result. It should show user a Success message.
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